

ALGORITHM – HIERARCHICAL INHERITANCE

Inheritance Structure: $A \rightarrow B$ and $A \rightarrow C$

1. Start.
2. Create class A with variables **a = 100**, **b = 10** and method **sum()** to calculate $a + b$.
3. Create class B that extends class A. Set **a = 10**, **b = 50** and create method **sub()** to calculate $b - a$.
4. Create class C that extends class A. Set **a = 25**, **b = 25** and create method **mul()** to calculate $a \times b$.
5. In the main() method, create an object **tv** of class C.
6. Call **tv.mul()**. Calculate 25×25 and display **625**.
7. Call **tv.sum()**. Calculate $100 + 10$ and display **110**.
8. Display the results.
9. Stop.

Quick Flow

START → Create A → Create B & C from A → Create C object → mul() → sum() → Display output → **STOP**

Output

multiplication is 625

sum is 110

Mnemonic: $A \rightarrow B$, $A \rightarrow C$ = Hierarchical Inheritance